Article

The scalar spectral constant of the quantum annulus

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Abstract

Fix $R > 1$. The quantum annulus consists of the invertible Hilbert-space operators A such that $\|A\|, \|A^{-1}\| \leq R$. We prove that its scalar spectral constant for the closed annulus $\{z \in \mathbb{C} : R^{-1} \leq |z| \leq R\}$ is exactly 2. The upper bound is first established for matrices. A shifted annular double-layer density gives a unital positive map of total mass 2. Applied to the Cayley transforms of a bounded scalar function, this map produces a matrix-valued positive-real completion whose defect lies in the adjoint algebra of an auxiliary simple-spectrum matrix. A direct sampling argument for the matrix Herglotz kernel then gives the sharp norm bound. Approximation removes the simple-spectrum and strictness assumptions. For arbitrary Hilbert spaces, a quantum-annulus boundary dilation is combined with residual finite dimensionality of the full unital free product $C(\mathbb{T}) *_{\mathbb{C}} \mathbb{C}^2$. Finite cyclic weighted shifts give the matching lower bound. The result concerns scalar functions; no assertion about the complete spectral constant is made.

Keywords: quantum annulus; spectral set; double-layer potential; matrix-valued Herglotz kernel; residually finite-dimensional C-star algebra

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1. Introduction

For $R > 1$, write

$$\mathbb{A}_R = \{z \in \mathbb{C} : R^{-1} < |z| < R\}, \qquad \overline{\mathbb{A}}_R = \{z \in \mathbb{C} : R^{-1} \leq |z| \leq R\}. \tag{1}$$

For a complex Hilbert space $\mathcal{H}$, the corresponding quantum annulus is

$$\mathbb{QA}_R(\mathcal{H}) = \Big\{A \in \mathcal{B}(\mathcal{H}) : A \text{ is invertible, } \|A\| \leq R,\ \|A^{-1}\| \leq R\Big\}. \tag{2}$$

Every A in this class has spectrum in $\overline{\mathbb{A}}_R$. Let $K(R)$ be the least number such that

$$\|f(A)\| \leq K(R) \max_{z \in \overline{\mathbb{A}}_R} |f(z)| \tag{3}$$

for every Hilbert space $\mathcal{H}$, every $A \in \mathbb{QA}_R(\mathcal{H})$, and every rational function f with no pole on $\overline{\mathbb{A}}_R$.

Crouzeix and Greenbaum proved the dimension-free upper bound $K(R) \leq 1 + \sqrt{2}$ by an annular double-layer argument [1, Section 5]. Tsikalas proved the lower bound $K(R) \geq 2$ for every $R > 1$[2, Theorem 1.1]. The annular double-layer map and its associated operator classes were subsequently analyzed by Jury and Tsikalas [3]. Our main result closes the gap.

Theorem 1.1 (Main theorem)**.** *For every $R > 1$,*

$$K(R) = 2. \tag{4}$$

Equivalently, if $\mathcal{H}$ is a complex Hilbert space, $A \in \mathbb{QA}_R(\mathcal{H})$, and f is rational with no pole on $\overline{\mathbb{A}}_R$, then

$$\|f(A)\| \leq 2 \max_{z \in \overline{\mathbb{A}}_R} |f(z)|, \tag{5}$$

and the factor 2 cannot be decreased.

The proof has two upper-bound stages. In finite dimensions, positivity of a shifted annular double-layer density supplies the positive-real completion needed in Theorem 2.2. For general Hilbert spaces, the boundary dilation of McCullough and Pascoe [4, Theorem 1.1(c)] reduces Laurent polynomials to boundary operators; the free-product theorem of Exel and Loring [5, Theorem 3.2] then reduces their norms to finite dimensions. Section 6 gives a finite-dimensional lower-bound sequence.

We use I for the identity operator of the size dictated by context, X^* for the adjoint, and $\operatorname{Re} X = (X + X^*)/2$. For Hermitian matrices X, Y, the notation $X \preceq Y$ means that $Y - X$ is positive semidefinite. The unital algebra generated by an operator X is denoted by $\operatorname{alg}(X)$.

2. A positive-real completion theorem

We begin with the finite-dimensional mechanism that produces the constant 2.

Lemma 2.1 (Matrix Herglotz kernel)**.** *Let $F\colon \mathbb{D} \to \mathbb{M}_n(\mathbb{C})$ be analytic and suppose that* $\operatorname{Re} F(w) \succeq 0$ *for $w \in \mathbb{D}$. Then*

$$\mathcal{L}_F(w, z) = \frac{F(w) + F(z)^*}{1 - w\bar{z}} \tag{6}$$

is a positive kernel: for arbitrary $w_0, \ldots, w_m \in \mathbb{D}$ and $\xi_0, \ldots, \xi_m \in \mathbb{C}^n$,

$$\sum_{i,j=0}^{m} \xi_i^* \mathcal{L}_F(w_i, w_j) \xi_j \geq 0. \tag{7}$$

Proof. Set

$$\eta(\zeta) = \sum_{j=0}^{m} \frac{\xi_j}{1 - \overline{w_j}\zeta}, \qquad |\zeta| = 1.$$

For $0 \leq r < 1$, entrywise integration of the power series of F and of the two scalar Cauchy kernels gives, for each i, j,

$$\frac{1}{2\pi} \int_0^{2\pi} \frac{F(re^{\mathrm{i}t})}{(1 - w_i e^{-\mathrm{i}t})(1 - \overline{w_j} e^{\mathrm{i}t})}\, dt = \frac{F(rw_i)}{1 - w_i \overline{w_j}},$$

$$\frac{1}{2\pi} \int_0^{2\pi} \frac{F(re^{\mathrm{i}t})^*}{(1 - w_i e^{-\mathrm{i}t})(1 - \overline{w_j} e^{\mathrm{i}t})}\, dt = \frac{F(rw_j)^*}{1 - w_i \overline{w_j}}.$$

After summing against the vectors ξ_i, ξ_j, these identities yield

$$\begin{aligned} &\frac{1}{2\pi} \int_0^{2\pi} \eta(e^{\mathrm{i}t})^* \, 2 \operatorname{Re} F(re^{\mathrm{i}t}) \, \eta(e^{\mathrm{i}t})\, dt \\ &\qquad = \sum_{i,j=0}^{m} \xi_i^* \frac{F(rw_i) + F(rw_j)^*}{1 - w_i \overline{w_j}} \xi_j. \end{aligned} \tag{8}$$

The left side is nonnegative. Letting r increase to 1 proves (7). □

Theorem 2.2 (Positive-real completion)**.** *Suppose*

$$B = S \operatorname{diag}(\beta_1, \ldots, \beta_n) S^{-1}, \qquad T = S \Lambda S^{-1}, \qquad \Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n), \tag{9}$$

where S is invertible, the numbers $\beta_1, \dots, \beta_n$ are distinct, and $|\lambda_i| \le 1$. The numbers λ_i need not be distinct. Let $H\colon \mathbb{D} \to \mathbb{M}_n(\mathbb{C})$ be analytic and satisfy

$$H(0) = I, \qquad \operatorname{Re} H(w) \succeq 0 \quad (w \in \mathbb{D}), \tag{10}$$

and

$$H(w) - (I - wT)^{-1} \in \operatorname{alg}(B^*) \quad (w \in \mathbb{D}). \tag{11}$$

Then $\|T\| \le 2$.

Proof. Put $\Gamma = S^*S \succ 0$. Since the eigenvalues of B are distinct, polynomial interpolation gives

$$\operatorname{alg}(B^*) = \{(S^{-1})^*\Delta S^* : \Delta \text{ is diagonal}\}. \tag{12}$$

Consequently, there is a diagonal analytic matrix $\Theta(w) = \operatorname{diag}(\theta_1(w), \dots, \theta_n(w))$, with $\Theta(0) = 0$, such that

$$\widetilde{H}(w) := S^*H(w)S = \Gamma(I - w\Lambda)^{-1} + \Theta(w)\Gamma. \tag{13}$$

The real part of $\widetilde{H}$ is positive, so Lemma 2.1 applies to $\widetilde{H}$.

Define Hermitian matrices P, Q, Y by

$$P_{ij} = \frac{\Gamma_{ij}}{1 - \overline{\lambda_i}\lambda_j/4}, \qquad Q_{ij} = \frac{\Gamma_{ij}}{1 - \overline{\lambda_i}\lambda_j/2}, \qquad Y = Q - P. \tag{14}$$

Fix $x = (x_1, \dots, x_n)^T \in \mathbb{C}^n$. In the kernel inequality, use

$$w_i = \frac{\overline{\lambda_i}}{2}, \qquad \xi_i = x_i e_i \quad (1 \le i \le n), \tag{15}$$

and add the sample $w_0 = 0$ with vector

$$\xi_0 = v = -\Gamma^{-1}Px. \tag{16}$$

Repeated sample points cause no difficulty.

The contribution of the term $\Theta(w)\Gamma$ in (13) to the sampled kernel quadratic form equals

$$2\operatorname{Re}\sum_{i=1}^{n} \overline{x_i}\theta_i(w_i)\left((\Gamma v)_i + \sum_{j=1}^{n} \frac{\Gamma_{ij}x_j}{1 - w_i\overline{w_j}}\right). \tag{17}$$

The expression in parentheses is $(\Gamma v + Px)_i = 0$, so the unknown term cancels exactly.

For the known term $\Gamma(I - w\Lambda)^{-1}$, compression of the kernel block between w_i and w_j by e_i and e_j gives

$$\frac{2\Gamma_{ij}}{(1 - \overline{\lambda_i}\lambda_j/2)(1 - \overline{\lambda_i}\lambda_j/4)} = (4Q - 2P)_{ij}. \tag{18}$$

Here the equality on the right follows from

$$\frac{2}{(1 - a/2)(1 - a/4)} = \frac{4}{1 - a/2} - \frac{2}{1 - a/4}.$$

Likewise, direct substitution of $w_0 = 0$ gives

$$\left[e_i^*\left(\Gamma(I - w_i\Lambda)^{-1} + \Gamma\right)e_j\right]_{i,j} = \Gamma + Q, \qquad \Gamma(I - 0\Lambda)^{-1} + \Gamma = 2\Gamma \tag{19}$$

for the known term. Kernel positivity, evaluated on the graph $v = -\Gamma^{-1}Px$, therefore gives

$$\begin{aligned} 0 &\le \begin{pmatrix} x \\ v \end{pmatrix}^* \begin{pmatrix} 4Q - 2P & \Gamma + Q \\ \Gamma + Q & 2\Gamma \end{pmatrix} \begin{pmatrix} x \\ v \end{pmatrix} \\ &= x^*\big(4Y - Y\Gamma^{-1}P - P\Gamma^{-1}Y\big)x. \end{aligned} \tag{20}$$

Since x is arbitrary,

$$4Y - Y\Gamma^{-1}P - P\Gamma^{-1}Y \succeq 0. \tag{21}$$

Balance the diagonalization by setting

$$\widetilde{T} = \Gamma^{1/2}\Lambda\Gamma^{-1/2}, \qquad \widehat{P} = \Gamma^{-1/2}P\Gamma^{-1/2}, \qquad \widehat{Y} = \Gamma^{-1/2}Y\Gamma^{-1/2}. \tag{22}$$

Expanding the scalar denominators in (14) gives the norm-convergent series

$$\widehat{P} = \sum_{k=0}^{\infty} 4^{-k}\widetilde{T}^{*k}\widetilde{T}^k, \tag{23}$$

$$\widehat{Y} = \sum_{k=1}^{\infty} (2^{-k} - 4^{-k})\widetilde{T}^{*k}\widetilde{T}^k \succeq 0. \tag{24}$$

Indeed, the sequence $(\widetilde{T}^k)$ is bounded because $\widetilde{T}^k = \Gamma^{1/2}\Lambda^k\Gamma^{-1/2}$. Congruence of (21) by $\Gamma^{-1/2}$ yields

$$4\widehat{Y} - \widehat{Y}\widehat{P} - \widehat{P}\widehat{Y} \succeq 0. \tag{25}$$

Let e be an eigenvector of $\widehat{P}$, with $\widehat{P}e = \alpha e$. If $\alpha > 2$, then (25) gives

$$0 \le 2(2-\alpha)e^*\widehat{Y}e.$$

Thus $e^*\widehat{Y}e = 0$. The $k = 1$ term of (24) is $\widetilde{T}^*\widetilde{T}/4$, so $\widetilde{T}e = 0$. Equation (23) then gives $\widehat{P}e = e$, contradicting $\alpha > 2$. Hence

$$I \preceq \widehat{P} \preceq 2I. \tag{26}$$

The $k = 1$ term of (23) now gives

$$\frac{1}{4}\widetilde{T}^*\widetilde{T} \preceq \widehat{P} - I \preceq I,$$

and therefore $\|\widetilde{T}\| \le 2$. Finally, the polar decomposition $S = U_S\Gamma^{1/2}$ gives $T = U_S\widetilde{T}U_S^*$, so $\|T\| \le 2$. $\square$

3. The annular double-layer completion

Let $\mathcal{A}(\mathbb{A}_R)$ denote the functions holomorphic on $\mathbb{A}_R$ and continuous on $\overline{\mathbb{A}}_R$. We now construct the completion in Theorem 2.2. The shifted density below is the annular double-layer density underlying the contractive map in [1, Section 5]; see also [3, Theorem 3.12]. We include the factorization, mass, and cancellation identities needed here.

Let $B \in \mathbb{M}_n(\mathbb{C})$ satisfy

$$\|B\| < R, \qquad \|B^{-1}\| < R. \tag{27}$$

Orient the outer circle of $\partial\mathbb{A}_R$ counterclockwise and the inner circle clockwise. On either component, let $z = z(s)$ be an oriented arclength parametrization. Define

$$\mu_B(z) = \frac{1}{2\pi \mathrm{i}}\Big[z'(s)(zI - B)^{-1} - \overline{z'(s)}(\bar{z}I - B^*)^{-1}\Big], \tag{28}$$

$$\kappa_R(z) = \frac{1}{2\pi \mathrm{i}}\frac{z'(s)}{z}, \qquad D_B(z) = \mu_B(z) - \kappa_R(z)I. \tag{29}$$

Both $\mu_B(z)$ and $D_B(z)$ are self-adjoint.

Lemma 3.1 (Positive shifted density)**.** *Under* (27),

$$D_B(z) \succeq 0 \quad (z \in \partial\mathbb{A}_R), \qquad \int_{\partial\mathbb{A}_R} D_B(z)\, ds = 2I. \tag{30}$$

Moreover, for every $h \in \mathcal{A}(\mathbb{A}_R)$,

$$\int_{\partial\mathbb{A}_R} h(z)\kappa_R(z)\, ds = 0. \tag{31}$$

Proof. On the outer circle, write $z = Re^{\mathrm{i}t}$. Then $\kappa_R(z) = 1/(2\pi R)$, and direct multiplication by $zI - B$ and $\bar{z}I - B^*$ gives

$$2\pi R D_B(z) = (zI - B)^{-1}(R^2 I - BB^*)(\bar{z}I - B^*)^{-1} \succeq 0. \tag{32}$$

On the inner circle, put $\rho = R^{-1}$ and use the clockwise parametrization $z = \rho e^{-\mathrm{i}t}$. Now $\kappa_R(z) = -1/(2\pi\rho)$ and

$$2\pi\rho D_B(z) = (zI - B)^{-1}(BB^* - \rho^2 I)(\bar{z}I - B^*)^{-1} \succeq 0. \tag{33}$$

The middle factors are positive because $BB^* \preceq R^2 I$ and $BB^* \succeq R^{-2} I$.

The matrix Cauchy formula gives

$$\frac{1}{2\pi\mathrm{i}} \int_{\partial\mathbb{A}_R} z'(s)(zI - B)^{-1}\, ds = \frac{1}{2\pi\mathrm{i}} \int_{\partial\mathbb{A}_R} (zI - B)^{-1}\, dz = I. \tag{34}$$

Under the change of variable $\zeta = \bar{z}$, the conjugate of the oriented annular boundary has the reverse orientation. Therefore

$$-\frac{1}{2\pi\mathrm{i}} \int_{\partial\mathbb{A}_R} \overline{z'(s)}(\bar{z}I - B^*)^{-1}\, ds = -\frac{1}{2\pi\mathrm{i}} \int_{-\partial\mathbb{A}_R} (\zeta I - B^*)^{-1}\, d\zeta = I. \tag{35}$$

Adding (34) and (35) yields

$$\int_{\partial\mathbb{A}_R} \mu_B(z)\, ds = 2I. \tag{36}$$

The winding numbers of the outer and inner circles are 1 and -1, respectively, so

$$\int_{\partial\mathbb{A}_R} \kappa_R(z)\, ds = \frac{1}{2\pi\mathrm{i}} \int_{\partial\mathbb{A}_R} \frac{dz}{z} = 0.$$

This proves the mass identity. Finally,

$$\int_{\partial\mathbb{A}_R} h(z)\kappa_R(z)\, ds = \frac{1}{2\pi\mathrm{i}} \int_{\partial\mathbb{A}_R} \frac{h(z)}{z}\, dz = 0$$

by Cauchy's theorem on the annulus. □

Define

$$\Phi_B(\varphi) = \frac{1}{2} \int_{\partial\mathbb{A}_R} \varphi(z) D_B(z)\, ds, \qquad \varphi \in C(\partial\mathbb{A}_R). \tag{37}$$

Lemma 3.1 makes Φ_B a unital positive map; in particular, it is bounded and $*$-preserving. For $h \in \mathcal{A}(\mathbb{A}_R)$, define

$$(\mathcal{C}\bar{h})(\zeta) = \frac{1}{2\pi\mathrm{i}} \int_{\partial\mathbb{A}_R} \frac{\overline{h(z)}}{z - \zeta}\, dz, \qquad \zeta \in \mathbb{A}_R. \tag{38}$$

This is holomorphic in $\mathbb{A}_R$. Splitting the two terms of μ_B and using (31) gives

$$\begin{aligned}\int_{\partial\mathbb{A}_R} h(z)D_B(z)\,ds &= \frac{1}{2\pi\mathrm{i}}\int_{\partial\mathbb{A}_R} h(z)(zI-B)^{-1}\,dz\\ &\quad -\frac{1}{2\pi\mathrm{i}}\int_{\partial\mathbb{A}_R} h(z)(\bar{z}I-B^*)^{-1}\,d\bar{z}\\ &= h(B)+\left[\frac{1}{2\pi\mathrm{i}}\int_{\partial\mathbb{A}_R} \overline{h(z)}(zI-B)^{-1}\,dz\right]^*.\end{aligned}$$

The bracket is $(\mathcal{C}\bar{h})(B)$ by (38) and the holomorphic functional calculus. Since the left side is $2\Phi_B(h)$, we obtain

$$2\Phi_B(h) = h(B) + (\mathcal{C}\bar{h})(B)^*. \tag{39}$$

Proposition 3.2 (Annular Cayley completion)**.** *Suppose that $B\in\mathbb{M}_n(\mathbb{C})$ satisfies (27) and has n distinct eigenvalues. If $f\in\mathcal{A}(\mathbb{A}_R)$, then*

$$\|f(B)\| \le 2\max_{z\in\overline{\mathbb{A}}_R}|f(z)|. \tag{40}$$

Proof. Put $M=\max_{z\in\overline{\mathbb{A}}_R}|f(z)|$. If $M=0$, then $f=0$ and the result is immediate. Replacing f by f/M, we may assume that $M\le 1$. Put $T=f(B)$. For $w\in\mathbb{D}$, define

$$c_w(z)=\frac{1+wf(z)}{1-wf(z)}, \qquad H(w)=\Phi_B(c_w). \tag{41}$$

The series $c_w = 1+2\sum_{m\ge1} w^m f^m$ converges locally uniformly in the supremum norm on $\overline{\mathbb{A}}_R$. Thus H is analytic and $H(0)=I$. Since

$$\operatorname{Re} c_w(z) = \frac{1-|wf(z)|^2}{|1-wf(z)|^2}\ge 0,$$

positivity and $*$-preservation of Φ_B give $\operatorname{Re} H(w)\succeq 0$.

Equation (39) gives

$$2H(w) = c_w(B) + (\mathcal{C}\overline{c_w})(B)^*. \tag{42}$$

Since

$$c_w(B) = (I+wT)(I-wT)^{-1} = 2(I-wT)^{-1}-I,$$

we obtain

$$H(w)-(I-wT)^{-1} = \frac{1}{2}((\mathcal{C}\overline{c_w})(B)^*-I)\in \operatorname{alg}(B^*). \tag{43}$$

Indeed, a holomorphic function of a finite matrix is a polynomial in that matrix, by interpolation with multiplicities, and hence belongs to its generated unital algebra.

Choose an eigenbasis $B=S\operatorname{diag}(\beta_1,\dots,\beta_n)S^{-1}$. Then

$$T=S\operatorname{diag}(f(\beta_1),\dots,f(\beta_n))S^{-1}, \qquad |f(\beta_i)|\le 1.$$

Theorem 2.2, applied to (43), yields $\|f(B)\|\le 2$ under the normalization $M\le 1$. Scaling back by M proves (40). □

4. The finite-dimensional estimate

Theorem 4.1 (Finite dimensions)**.** *Let $A\in\mathbb{M}_n(\mathbb{C})$ be invertible and satisfy $\|A\|,\|A^{-1}\|\le R$. If f is rational and has no pole on $\overline{\mathbb{A}}_R$, then*

$$\|f(A)\| \le 2\max_{z\in\overline{\mathbb{A}}_R}|f(z)|. \tag{44}$$

Proof. Write the polar decomposition as $A = U_A|A|$. Because A is invertible, U_A is unitary and

$$R^{-1}I \preceq |A| \preceq RI. \tag{45}$$

For $0 < \varepsilon < 1$, put

$$A_\varepsilon = U_A\big((1-\varepsilon)|A| + \varepsilon I\big). \tag{46}$$

Since $R > 1$, (45) implies

$$\|A_\varepsilon\| < R, \qquad \|A_\varepsilon^{-1}\| < R, \qquad A_\varepsilon \longrightarrow A. \tag{47}$$

Indeed, the spectrum of the positive factor in (46) lies in the open interval (R^{-1}, R).

The strict inequalities in (47) persist under sufficiently small matrix perturbations. Matrices with distinct eigenvalues are dense: the discriminant of the characteristic polynomial is a nonzero polynomial in the matrix entries, so its zero set has empty interior. Hence, for each ε, there are simple-spectrum matrices $B_{\varepsilon,k} \to A_\varepsilon$ that satisfy (27). Proposition 3.2 gives

$$\|f(B_{\varepsilon,k})\| \le 2\max_{z\in\overline{\mathbb{A}}_R}|f(z)|.$$

First let $k \to \infty$ and then let $\varepsilon \downarrow 0$. Continuity of rational matrix evaluation proves (44). $\square$

5. Passage to arbitrary Hilbert spaces

We use two external structural theorems in this section. In our normalization, the boundary-dilation theorem of McCullough and Pascoe [4, Theorem 1.1 and Proposition 3.2] says that every $A \in \mathbb{QA}_R(\mathcal{H})$ has a Laurent dilation to an invertible operator $J \in \mathcal{B}(\mathcal{L})$: there is an isometry $V\colon \mathcal{H} \to \mathcal{L}$ such that

$$A^m = V^*J^mV \quad (m \in \mathbb{Z}), \tag{48}$$

and

$$(R^2 + R^{-2})I - J^*J - (J^*J)^{-1} = 0. \tag{49}$$

The second input is the theorem of Exel and Loring [5, Theorem 3.2]: the full unital free product of two unital residually finite-dimensional C^*-algebras is residually finite-dimensional.

Set

$$\mathfrak{F} = C(\mathbb{T}) *_{\mathbb{C}} \mathbb{C}^2. \tag{50}$$

Let $u_0 \in \mathfrak{F}$ be the canonical unitary from $C(\mathbb{T})$ and let $p_0 \in \mathfrak{F}$ be the canonical projection from $\mathbb{C}^2$. Define

$$j_R = u_0\big(Rp_0 + R^{-1}(1-p_0)\big). \tag{51}$$

Its inverse is

$$j_R^{-1} = \big(R^{-1}p_0 + R(1-p_0)\big)u_0^*. \tag{52}$$

The algebra $C(\mathbb{T})$ is residually finite-dimensional because point evaluations separate its elements, and $\mathbb{C}^2$ is finite-dimensional. Thus $\mathfrak{F}$ is residually finite-dimensional. Equivalently,

$$\|x\| = \sup_\pi \|\pi(x)\| \qquad (x \in \mathfrak{F}), \tag{53}$$

where π ranges over finite-dimensional unital $*$-representations. Indeed, Exel and Loring's characterization [5, Theorem 2.4] makes their direct sum faithful and hence isometric.

Lemma 5.1 (Boundary reduction)**.** *If J satisfies (49), then every Laurent polynomial q satisfies*

$$\|q(J)\| \le \sup_{n\ge1}\ \sup_{B\in\mathbb{QA}_R(\mathbb{C}^n)} \|q(B)\|. \tag{54}$$

Proof. Put $Z = J^*J$. Multiplication of (49) by Z gives

$$(Z - R^2 I)(Z - R^{-2} I) = 0.$$

The spectral theorem therefore shows that the spectrum of Z is contained in $\{R^2, R^{-2}\}$. Hence

$$Z = R^2 E + R^{-2}(I - E) \tag{55}$$

for a projection E. The polar decomposition of J is therefore

$$J = U_J\big(RE + R^{-1}(I - E)\big) \tag{56}$$

with U_J unitary. The universal property of (50) gives a unital $*$-representation $\rho_J \colon \mathfrak{F} \to \mathcal{B}(\mathcal{L})$ such that $\rho_J(j_R) = J$. Thus

$$\|q(J)\| \le \|q(j_R)\|.$$

By (53), the last norm is the supremum of $\|q(\pi(j_R))\|$ over finite-dimensional unital $*$-representations π. Each matrix

$$\pi(j_R) = \pi(u_0)\big(R\pi(p_0) + R^{-1}(I - \pi(p_0))\big)$$

has norm at most R. Its inverse is

$$\big(R^{-1}\pi(p_0) + R(I - \pi(p_0))\big)\pi(u_0)^*,$$

which also has norm at most R. Thus $\pi(j_R)$ belongs to the finite-dimensional quantum annulus, proving (54). $\square$

Proof of the upper bound in Theorem 1.1. Let $A \in \mathbb{QA}_R(\mathcal{H})$ and let J, V be supplied by (48)–(49). If $q(z) = \sum_{m=-N}^{N} c_m z^m$ is a Laurent polynomial, then $q(A) = V^* q(J) V$. Lemma 5.1 and Theorem 4.1 give

$$\begin{aligned} \|q(A)\| &\le \|q(J)\| \\ &\le \sup_{n \ge 1} \sup_{B \in \mathbb{QA}_R(\mathbb{C}^n)} \|q(B)\| \le 2 \max_{z \in \overline{\mathbb{A}}_R} |q(z)|. \end{aligned} \tag{57}$$

Now let f be rational with no pole on $\overline{\mathbb{A}}_R$. There are radii $r_0 < R^{-1} < R < r_1$ such that f is holomorphic on $r_0 < |z| < r_1$. Its Laurent series on this annulus has partial sums q_k satisfying

$$\max_{z \in \overline{\mathbb{A}}_R} |q_k(z) - f(z)| \longrightarrow 0. \tag{58}$$

The convergence also holds on two circles surrounding $\sigma(A)$ inside the larger annulus, so the Cauchy integral formula for the holomorphic functional calculus gives $q_k(A) \to f(A)$ in operator norm. Passing to the limit in (57) proves (5). $\square$

6. Sharpness

The lower bound is approached by finite cyclic weighted shifts. This is a finite-dimensional version of the weighted-shift construction in [2, Section 2].

For $n \ge 1$, let W_n act on $\mathbb{C}^{2n}$ with standard basis $e_0, \dots, e_{2n-1}$ by

$$W_n e_k = \omega_k e_{k+1 \pmod{2n}}, \qquad \omega_k = \begin{cases} R, & 0 \le k < n, \\ R^{-1}, & n \le k < 2n. \end{cases} \tag{59}$$

The cyclic permutation is unitary, so the singular values of W_n are the numbers ω_k. Consequently,

$$\|W_n\| = \|W_n^{-1}\| = R. \tag{60}$$

For the Laurent polynomial

$$g_n(z) = R^{-n}(z^n + z^{-n}), \tag{61}$$

the maximum-modulus principle, applied on the annulus, gives

$$\max_{z \in \overline{\mathbb{A}}_R} |g_n(z)| = 1 + R^{-2n}. \tag{62}$$

Indeed, on either boundary circle the triangle inequality gives this upper bound, and equality holds at a positive real boundary point. On the other hand, the two half-cycles in (59) give

$$W_n^n e_0 = W_n^{-n} e_0 = R^n e_n.$$

Hence

$$\|g_n(W_n)\| \geq 2. \tag{63}$$

It follows from (60)–(63) that

$$K(R) \geq \frac{2}{1 + R^{-2n}}.$$

Letting $n \to \infty$ proves $K(R) \geq 2$. Together with the upper bound, this completes the proof of Theorem 1.1.

Remark 6.1 (Scalar scope)**.** The theorem identifies the scalar spectral constant. It makes no assertion about the complete spectral constant.

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OpenAI ChatGPT assisted with checking the kernel-sampling calculation, the annular boundary factorizations, the source-to-claim matching, and the exposition and typesetting of this manuscript. The human author assumes full responsibility for the correctness of all mathematical statements and for the integrity and accuracy of all citations.

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